CURBCOUNT · FEASIBILITY ASSESSMENT

CurbCount feasibility assessment

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Opened in the kit 1.9.1 upgrade

CurbCount feasibility assessment

The thermal array counter is feasible to build and mount on paper at the 275 USD target (274.00 USD), and its main weakness is counting accuracy, not construction; the radar option is feasible in principle but not shown to fit the cost or the rules. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The thermal array design is constructable on paper: 127 parts modelled at D3 with real stock, fasteners and clearances, interference-free; wind, twist, coverage, energy and radio close in CAL-001	High	The tracker's frame-count margin for cars is 4.2 against 4 needed (PR-001); sunlit pavement hides people for part of a warm day (PR-008)	STM S1 to S7, CAL-001, cad/fidelity.json
Materials	Aluminium sections and sheet, printed ASA, a polycarbonate FieldNode enclosure, stainless fasteners and bands: all stock	Medium	HDPE window film grade for UV is unchosen (S18); aluminium against stainless and a steel pole needs a coating or isolating washers (S19)	STM S18, S19; BOM
Manufacturing and assembly	Garage-buildable: cut and drill aluminium, print one housing, assemble with hand tools; 14 BOM lines; install by two people in about 59 minutes	High	Many holes are drilled to a template; the build plan (BLD-001) gives the order and the jigs	BLD-001, CAL-001 H6
Power and controls	One LiFePO4 cell and a 6 W panel give 6.22 days of autonomy and 4.4 Wh/day against 2.47; the tracker takes about 2 % of the processor	Medium	Charging stops below about 0 degrees C air (S8, open decision O1); cycle count of the tracker is an estimate until profiled (S16)	STM S6 to S8, S16; CAL-001 E, F
Supply of parts	Every line is a catalogue part or a stock section; the array breakout is the one costly line (65.00 USD, SparkFun class)	Medium	The BOM total sits 1.00 USD under the 275 USD target on indicative prices; the sensor price was checked on 2026-10-31 (64.95 USD) and the rest is unchecked	BOM, MKT
Sensing and data (the risk the design rests on)	The principle holds (S1, S2) but accuracy is not shown until counts are compared with manual counts	Low	PR-001, PR-002 and PR-008 are at risk on paper; a field comparison is TRL 4 work and on hold	CAL-001 C, D; PRS
Option B (radar head)	Modelled at D3 (128 parts); energy closes only with a duty-cycled radar, a 20 W panel and two cells	Low	Module price and 60 GHz rules unknown (G9); tracker burst rate to be profiled; 60 GHz hardware is harder to build in a garage	STM S21 to S24; OPT

Ratings: High, Medium, Low.